Engineering Chemistry · Unit_04 · Functional materials and green chemistry

DRAFT — UE26CY151A · Unit_04 · Functional materials and green chemistry · Polymers: Introduction, types of polymerizations, definition of glass transition temperature

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

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M1

Polymers: Introduction, types of polymerizations, definition of glass transition temperature

1 Polymers: Introduction, types of polymerizations, definition of glass transition temperature

Study focus: Polymers: Introduction, types of polymerizations, definition of glass transition temperature. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000054 · EV0000055

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we'll start with a new module on functional materials.

Key details

- In that module we are going to study about polymers, the types of polymerization, structure property relationship, commercial polymers will take up butile rubber, biodegradable polymers, epoxy resin, polymer composites, cablar and carbon fiber.
- Conducting polymers and we will look at molecular weight of polymers.
- Then we'll also study organic LEDs.
- First, we will take up some definitions.
- A giant molecule or a macromolecule, which is formed by repeated joining of several simple molecules.
- If you look at the picture here, we can see some small molecules over here and they have been joined together by a process which is called polymerization to give this long chain and this chain of these simple molecules is known as polymer.
- What we have to know is between each of these small simple molecules the bond is a covalent bond now these polymer chains they could be linear they could be branched like this or they there could be a cross link network type of polymer.
- What is a monoma then?
- A simple molecule having two or more binding sites which can be linked together to form a giant molecule.
- When we say simple molecule, do all simple molecules classify as small modes?

- What is the structure of ethylene?
- CH2, double bond, CH2.
- What is the criteria?
- There should be two or more binding sites.
- Yes, we find that there are two binding sites across the double bond.
- The double bond can open and we get two carbons wherein the bonding can happen.
- Now here we have just one functional group, so therefore it cannot bind with several molecules.
- It can maximum m bind with one more ovastic acid and you just end up with a dimer.
- You definitely cannot get a polymer.
- No, there are no binding sites at all.
- What is ethylene glycol?
- Two OH groups, yes, if the binding can happen on both sides, therefore we could get a polymer from here.
- What is adipic acid?
- We have COOH, and here we have CH2 four times.
- And again we have COOH.
- So definitely there are two carboxylas carboxylic acid groups here, therefore it will act as a monument.
- Amino acid we have NH2 and this.
- CH2 could be N times, and then we have the C O H.
- Yes, here also we have two different binding functional groups.
- Therefore, this also will qualify as a monomer.
- What is styrene?
- CH to CH and there is a benzene ring here.
- So phenyl group, therefore, yes.
- Where does the binding take place?
- The moment you have unsaturation then we know that it is possible that you.
- This double bond can open up and you have two binding sites and therefore this also qualifies as a monomer.
- The next definition is of polymerization.
- The process by which monomers of low molecular weight are converted to polymers of high molecular weight.
- You can see there are double bonds, so many ethene molecules are there, and when they are polymerized, you get a chain like this and this is polyethylene.
- One chain will be referred to as a polymer molecule.

Example or application from the lecture

- Let us understand what is a polymer.
- Let us take up some examples and understand this.
- Let us take this example of ethene.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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Core explanation · 05:00–09:59

The total number of functional groups or bonding sites which are present in a monoma.

Key details

- Double bond CH2.
- Just now we saw it is CH2OH.
- Therefore, again it is bifunctional.
- Glycerol is CH2OH.
- CHOH and then again CH2OH.
- Therefore, we know that there are three functional groups through which binding can happen.
- Just now we saw it has an NH2 group and then CH two it could be N number and then we have COOH.
- Again there are two functional groups, so it is again bifunctional in nature.
- Degree of polymerization.
- How would you find out the degree of polyzation?
- If the molecular weight of the polymer and you also know the molecular weight of the monomer which you have used, we can calculate the degree of polymerization, molecular weight of polymer divided by molecular weight of the monoma.
- What would be the use of knowing the degree of polymerization?
- We can determine the length of the polymer chain and we can also find out what is the molecular weight of the polymer.
- That would be n because how did you define degree of polymerization?
- The total number of repeat units.
- One is the addition polymerization and the other is the condensation polymerization.
- Addition polymerization would be self addition of unsaturated monomers to each other takes place without the elimination of any byproducts.
- chain polymerization and examples would be PVC polystyrene teflon polyethene.
- C x is some other.
- Entity except so it could be H also, it is just some substitute substitution substituted group over here.
- Of course the bonding is happening on both the directions and therefore we have n to n of these units.
- Just by changing the substituent to Cl, we got polyvinyl chloride, which is totally different from polyethylene in properties.
- What do we get?
- We get is all four are fluorine atoms now.
- Instead of hydrogens, we are using fluorine atoms and what we end up will be Teflon, which is very strong polymer.

Example or application from the lecture

- Let us take them up one by one.
- Let us see the reaction here.

- For example, if we have hydrogen as the substituent, we end up with polyethylene.

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Core explanation · 10:00–14:59

and has exceptional properties and it is known as wonder plastic.

Key details

- Then what we get is polystyrene.
- Just by changing the hydrogen atom.
- You could change one atom, you could change all the four, you could change any number of hydrogen atoms, each time it is going to give you a polymer with totally different properties.
- We can have linear or branched polymers.
- Self addition takes place very rapidly, as I mentioned, it's also known as chain polymerization.
- There's no elimination of byproducts.
- Addition polymerization can take place by.
- It could happen by free radical mechanism.
- It is going to generate a free radical.
- in the monomer and that is going to help in the propagation of the reaction.
- That is not the only mechanism which can be followed for addition polymerization, it is also possible to do by cationic mechanism.
- A cation is generated just like in the free radical mechanism, free radical is generated.
- Here we are going to generate a cation.
- It is a Lewis acid.
- Now if the mechanism is anionic mechanism, then obviously now we understand anion has to be generated and therefore we are going to use some other agent now and that would be an organometallic compound.
- Only known organic organometallic compounds have Wrignax reagents.
- Or it could also happen through coordination mechanism.
- What is a coordinate bond where the covalent bond where the electrons are shared by one of the atoms.
- If the polymerization is happening through this mechanism, what would be used here?
- This is happening catalyst when we say it always happens on the surface.
- The reaction is going to happen on the surface.
- Polymer which is produced is going to be a very stereorcular polymer.
- For addition polymerization, we should know that molecular weight is an integral multiple of that of the monoma.
- The reason being that there is no elimination of any byproducts.
- We can have different types.
- It could be a homopolymer where you have just one type of monoma being used, or it could be a copolymer where you will have.
- Two or more different types of monomers being used.

- we could again get different types of co-polymers.
- Depending on how these monomers are arranged.
- It's not necessary that they have to be alternate.

Example or application from the lecture

- Now another example would be if we replace the X by a C6H5.
- So if we want to carry out the polymerization through free radical mechanism the initiator used has to be some free radical generator and that example would be benzoyl peroxide.
- That would be example would be ALCl3.

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Core explanation · 15:00–19:59

Let us look at them.

Key details

- When we say a single monomer unit, then it's called homopolymer.
- When we have two or more different monomer units, it's called a copolymer.
- it is called a copolymer.
- It is alternating.
- You can see one is say M1, the other is M two, and similarly they are alternating throughout the chain.
- That would be an alternating co-polymer.
- Or it could be random means.
- The copoly the two monomer units are just appearing randomly.
- There's no particular arrangement that would be a random copoly mark.
- Then we can also have a block copolymer where we have a block of one type of monomer, and then we have a block of another type of monoma.
- Then again, this block.
- Is coming up and then we have another block.
- Talking about condensation, polymerization, it would be intermolecular condensation through functional group of the monomers with continuous elimination of byproducts.
- And common examples would be nylons, polyester, and phenol formaldehyde resins all these are condensation polymers.
- Here we have the adipic acid and we have hexamethylene diamine.
- Hexamethlene diamine, when we use that, there is an elimination of the so here is sorry this is hexamethylene diamine and this is the adipic acid we see that COOH group here so OH from here and the hydrogen from the hexamethylene diamine is being used so hydrogen of the amine group.
- The linkage here is an amide linkage.
- CONH group is known as the MI group.
- Adipic acid and elimination of water molecule is going to give you this bonding between the two molecules it will take place and this rip unit is going to be repeated throughout the

polymer chain.

- But what is the linkage we are getting here?
- We call it a polyester and the common name for using when you use terithalic acid and ethylene glycol is terylin.
- The as acids or alkali are being used as catalysts.
- In chain in chain polymerization, that is in addition polymerization we used initiators.
- We use the initiators like pre radicals, etcetera.
- Here it is cataly here we use a catalyst and this is also known as step polymerization.
- Chain polymerization is addition polymerization.
- Condensation polymerization is called step polymerization.
- Because the chain buildup happens stepwise, and there is continuous elimination of the byproducts.
- And one thing to note here is that linear branched or even cross-linked polymers are possible for condensation polymers.
- And of course, the molecular weight of the polymer is not an integral multiple.

Example or application from the lecture

- Now another example would be pterylene, where we have terithalic acid, again we have COOH here and ethylene glycol.

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Core explanation · 20:00–24:59

of the monoma because continuous elimination of these small molecules is happening.

Key details

- Differences between addition and condensation polymerization.
- Here there are small molecules are getting eliminated.
- Here initiators are required.
- Here we have catalyst and then linear or branch polymer are produced.
- Here we have cross linked also apart from linear and branched molecules.
- Weight is an integral multiple because there's no elimination of molecules, but small molecules here it is not an integral multiple of the monoma.
- And then monomers are unsaturated molecules are used as monomers and here they must have two active functional groups.
- All amorphous polybles when they are cooled below a certain temperature, they are found to become stiff, they become hard, brittle, and that state is known as the glassy state.
- But above the temperature, when it is heated, they become soft and flexible, and that is referred to as the rubbery state.
- This rubbery state when you heat it further then the.
- Polymer will start flowing, it gets into a viscous flow, and this temperature is known as the melting point of the polymer.
- One when it changes, you can see the graph here, one when it changes from the glass state.

- To the rubbery state.
- it when you heat it, when you're heating it inside, then it becomes into a viscous liquid and starts flowing, and when that change happens that is known as the melting point.
- Why is it changing like this?
- When we go to the very low temperature, I mean below the Tg, it is found that there is absolutely no segmental motion, neither the molecular motion.
- So it's like very stiff and therefore it behaves almost like a glass.
- Is that the glassy stint.
- You are giving energy to these molecules.
- But nevertheless, no molecular motion is here.
- So therefore only segmental motion.
- And this would be referred to as a rubbery state.
- But when you heat more, that means you will overcome all the forces of attraction between the polymer chains and the molecules now become absolutely free like this and therefore it change it becomes into a viscous liquid.
- If you want to fabricate a polymer or use it for a particular application, both TG and TM values are very important.
- We should know what are these values.
- So that means at room temperature we find that it is above the T G value and it is.
- Soft, whereas coming to polystyrene, we know polystyrene what it looks like, and there we see it is 100.
- So that means.
- At room temperature it is still in the glassy state.
- Where is the what would be the state at room temperature, and of course, when TG and TM you can fabricate and use this polymer for different applications.

Example or application from the lecture

- And that is given by Tg.

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Core explanation · 25:00–29:59

Properties of the polymer.

Key details

- In the next class, we are going to talk about the structure property relation between the in the in the polymers the relation between the structure and property of a polymer.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Polymers: Introduction, types of polymerizations, definition of glass transition temperature transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__6749a5eae673dbbc.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Polymers: Introduction, types of polymerizations, definition of glass transition temperature transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__6749a5eae673dbbc.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Polymers: Introduction, types of polymerizations, definition of glass transition temperature transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__6749a5eae673dbbc.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9986; only the JSON segmentation supplies the displayed timestamps.
Polymers: Introduction, types of polymerizations, definition of glass transition temperature transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/042_Polymers_Introduction_types_of_polymerizations_definition_of_glass_transition_temperature__6749a5eae673dbbc.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00014	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00015	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0058. Sidecar variants compare...
SRC00016	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00017	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0058. Sidecar variants compare...

C Revision and retrieval practice